

The SVD in Machine Learning

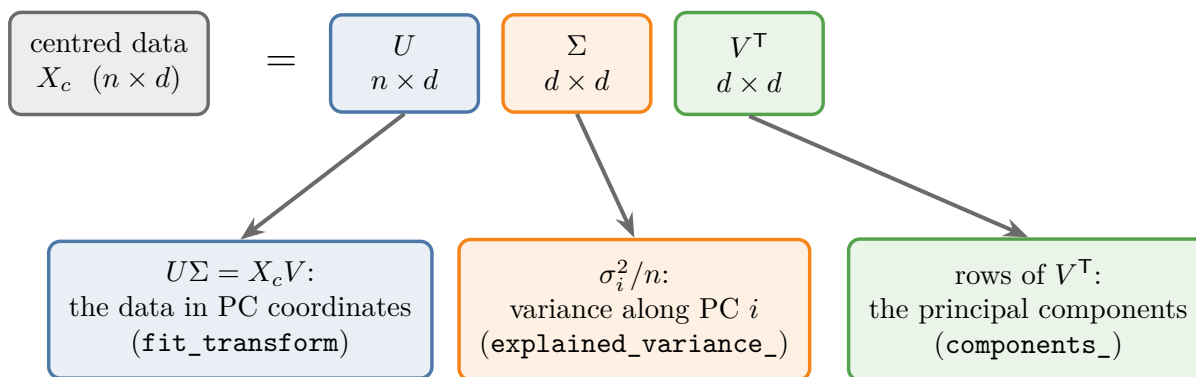
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1. Overview

Key point: PCA, latent semantic analysis, recommender systems and least-squares regression all run on the SVD: PCA reads the principal components off V , text and ratings models keep the top singular directions as hidden “topics”, and the pseudo-inverse solves regression even when the normal equation breaks.

This Note follows *Mathematics for Machine Learning* (Deisenroth, Faisal, Ong, 2020), §4.5 (Example 4.14, movie ratings, and the closing remarks on applications) and §4.6 (Example 4.15).



scikit-learn divides by $n - 1$ instead of n

Figure 1: PCA read off the thin SVD of the centred data

The first SVD Note, the [SVD geometry Note](#), defined $A = U\Sigma V^T$, and the [low-rank approximation Note](#) showed that the first k layers give the best rank- k approximation. This Note uses both in four ML tools:

- PCA computed through the SVD (Section 2 and Figure 1);
- latent semantic analysis of text (Section 3);
- a ratings matrix, the core of SVD-style recommenders (Section 4);
- the pseudo-inverse and least squares (Section 5).

2. PCA through the SVD

Key point: For centred data $X_c = U\Sigma V^T$, the rows of V^T are the principal components, σ_i^2/n are the variances along them, and $U\Sigma$ is the projected data. No covariance matrix is needed.

2.1 Why the SVD of the data gives PCA

Key point: The covariance matrix is $X_c^\top X_c/n$, and $X_c^\top X_c = V\Sigma^2V^\top$: its eigenvectors are the right singular vectors of the data.

The [PCA step by step Note](#) found the principal components as the eigenvectors of the covariance matrix, after centring each column. In matrix form, the covariance matrix of centred data with n rows is $X_c^\top X_c/n$: entry (j, k) is the sum of products of columns j and k , divided by n .

The [computing the SVD Note](#) (section 2.1) showed that $A^\top A = V\Sigma^\top \Sigma V^\top$ for any matrix. With $A = X_c$:

1. **In words:** the covariance matrix is the data's $A^\top A$ divided by n , so its eigenvectors are the right singular vectors of X_c and its eigenvalues are the squared singular values divided by n .
2. **Formula:**

$$C = \frac{1}{n}X_c^\top X_c = V \frac{\Sigma^2}{n} V^\top, \quad \text{variance along PC } i = \frac{\sigma_i^2}{n}, \quad Z = X_c V = U \Sigma$$

3. **Example:** for the 30 flats (rooms and washrooms) of the PCA Notes, the centred data has singular values 8.843 and 1.243:

$$\frac{8.843^2}{30} = 2.61, \quad \frac{1.243^2}{30} = 0.05$$

exactly the eigenvalues 2.61 and 0.05 found in the [PCA step by step Note](#) (section 4.3). The right singular vectors are $[0.707, 0.707]$ and $[-0.707, 0.707]$ (up to sign): PC1 at 45° , as there.

Figure 1 maps each part of the SVD to its PCA meaning:

- $V^\top$: its rows are the principal components, largest variance first. They are already sorted, because singular values come sorted.
- Σ : σ_i^2/n is the variance along PC i . The explained variance ratio is $\sigma_i^2/\sum_j \sigma_j^2$; for the flats, $8.843^2/(8.843^2 + 1.243^2) = 0.98$.
- $U\Sigma$: the projected data. Multiplying $X_c = U\Sigma V^\top$ by V on the right gives $X_c V = U\Sigma$, which is the projection $Z = XW^\top$ of the PCA Note with $W = V^\top$. The first flat, centred, lands at $[-0.639, 0.052]$.

2.2 Why compute it this way

Key point: The SVD works on the data directly, which is more accurate, and gives the projected data for free.

Forming $X_c^\top X_c$ squares the singular values, and small ones lose accuracy (the [computing the SVD Note](#), section 8). The SVD of X_c avoids that step. The squaring matters little for a few well-behaved columns, but it matters for data with nearly dependent columns.

Extra: What scikit-learn actually does (version 1.9). `PCA(svd_solver="auto")` picks a method from the shape of the data:

- few columns (at most 1,000) and at least 10 times as many rows: it eigen-decomposes the covariance matrix ("`covariance_eigh`", added in version 1.5), because a small $d \times d$ matrix is fastest there;
- otherwise, small data (no side above 500): a full SVD of the centred data ("`full`", LAPACK through SciPy);
- otherwise, when few components are wanted: a **randomized SVD**, which finds only the top k singular vectors.

So “scikit-learn’s PCA uses the SVD” was true for every input before version 1.5, and is still true for wide or large data. The 30 flats, and the 33,600 MNIST training images of 784 pixels in the [PCA on MNIST Note](#), now go through the covariance matrix. Each route gives the same components up to sign. PCA also divides by $n - 1$, so it reports 2.70 and 0.05 for the flats.

Python: PCA through the SVD.

```
Xc = P - P.mean(axis=0)           # P: rooms, washrooms
U, s, Vt = np.linalg.svd(Xc, full_matrices=False)
s ** 2 / len(P)                   # [2.607, 0.052]: variances
Vt                                # rows: PC1, PC2
Z = U * s                         # projected data, = Xc @ Vt.T
```

The Notebook builds `P` with the same random seed as the PCA Notes and checks the result against scikit-learn.

3. Latent semantic analysis

Key point: Keep the top k singular directions of a document-word count matrix. Documents that use different words for the same topic end up pointing the same way.

3.1 The problem with word counts

Key point: Bag-of-words vectors only see shared words, so two texts on the same topic with no word in common have cosine similarity 0.

The [vectors and feature vectors Note](#) (section 6) turned texts into bag-of-words count vectors, and the [dot product and cosine similarity Note](#) (section 6) compared them by angle. That only sees shared words. Take seven short documents (common words such as “a” and “and” removed):

	Document
d1	batsman hits a century
d2	bowler takes a wicket
d3	batsman and bowler: century, wicket, match
d4	chef cooks a curry

	Document
d5	rice and dal
d6	chef serves curry with rice and dal
d7	chef watches the match

d1 and d2 are both about cricket, but they share no word, so their cosine similarity is 0. The same holds for d4 and d5, both about food. A search for “batsman” would never find d2.

3.2 Topics from the SVD

Key point: Words that appear together load on the same singular vector; the first k singular directions act as k topics.

Stack the count vectors as the rows of a 7×14 matrix X (7 documents, 14 words). Its first two singular values are 2.73 and 2.64, and the rest are smaller. **Latent semantic analysis (LSA)** keeps the first k and describes each document by its coordinates along them.

1. **In words:** each document’s new coordinates are its row of $U_k \Sigma_k$: how strongly it uses each of the k word patterns $\mathbf{v}_1, \dots, \mathbf{v}_k$.
2. **Formula:**

$$Z = U_k \Sigma_k = X V_k$$

3. **Example:** with $k = 2$, d1 lands at $[0.635, 0.836]$ and d2 at exactly the same point. Their cosine similarity in this 2D topic space is 1.00, up from 0. d4 at $[0.932, -0.737]$ and d5 at $[0.596, -0.564]$ get 1.00 as well.

Figure 2 shows why. d3 contains all the cricket words together, so the SVD learns that “batsman”, “bowler”, “century” and “wicket” belong to one pattern. Through it, d1 and d2 are linked even though they share nothing directly. The food documents form the other pattern. d7 mixes “chef” with “match”, and lands in between, closer to food (cosine 0.86 with d4) than to cricket (0.49 with d1).

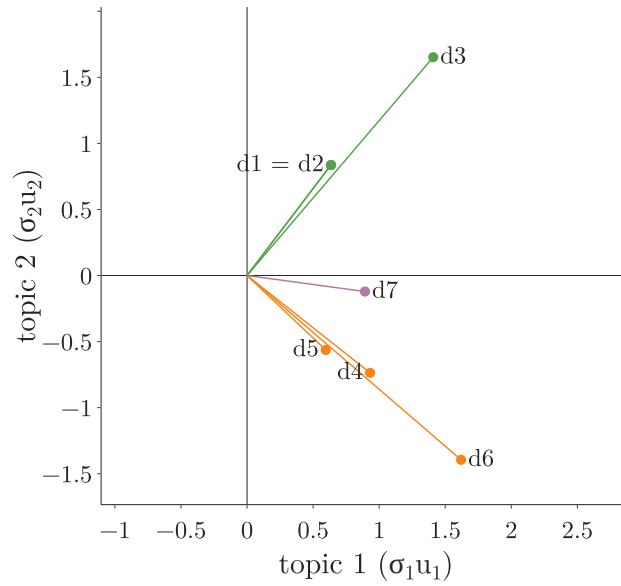


Figure 2: The seven documents in the 2D topic space of LSA: cricket (green), food (orange), and the mixed d7 (purple)

This is a rank- k approximation (the [low-rank approximation Note](#)) used for meaning rather than compression: dropping the small singular directions drops the accidents of word choice and keeps the shared themes. LSA was an early method of search engines for exactly this reason.

Python: TruncatedSVD keeps the top k singular directions.

```
from sklearn.decomposition import TruncatedSVD
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics.pairwise import cosine_similarity
```

```
X = CountVectorizer(stop_words="english").fit_transform(docs)
Z = TruncatedSVD(n_components=2, random_state=0).fit_transform(X)
cosine_similarity(Z)[0, 1]      # d1 vs d2: 1.0 (0 before)
```

Unlike PCA, TruncatedSVD does not centre the columns. Centring would turn a sparse count matrix (mostly zeros, stored compactly) into a dense one, which is impossible for a real vocabulary of 50,000 words.

4. Ratings and recommenders

Key point: In a viewers-by-films ratings matrix, the singular vectors become “kinds of viewer” and “kinds of film”; a rank-2 approximation already rebuilds the table.

4.1 A ratings matrix

Key point: Five viewers rate four films; two like action, two like romance, one likes everything.

Recommender systems work on a table of ratings: one row per viewer, one column per item. Figure 3 (left) shows five viewers rating four films from 1 to 5. Asha and Ben like the action films, Chitra and Dev the romance films, and Esha likes everything.

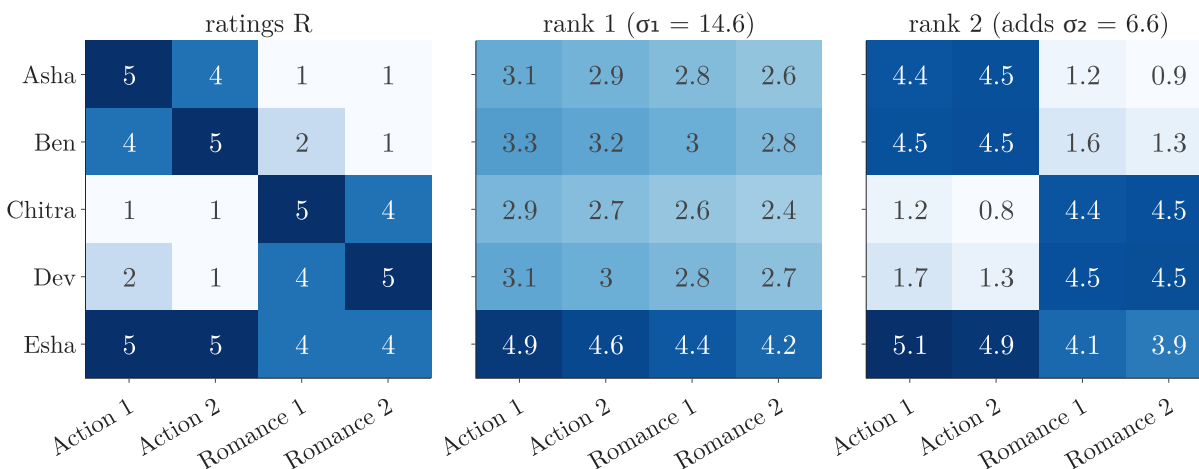


Figure 3: The ratings matrix and its rank-1 and rank-2 approximations

4.2 Reading the singular vectors

Key point: $\mathbf{v}_1$ is overall liking, $\mathbf{v}_2$ is action against romance; $\mathbf{u}_i$ says how much each viewer follows each pattern.

The singular values are 14.6, 6.6, 1.5 and 0.6. The first two dominate.

- $\mathbf{v}_1 = [-0.54, -0.51, -0.49, -0.46]$: all four films with the same sign and similar size. This is “how much a viewer likes films at all”. The rank-1 approximation (Figure 3, middle) gives each viewer roughly the same rating for every film: Esha high, the others near 3.
- $\mathbf{v}_2 = [0.44, 0.51, -0.50, -0.55]$: positive for the action films, negative for the romance films. This is a taste axis. The matching $\mathbf{u}_2 = [0.48, 0.42, -0.57, -0.51, 0.08]$ puts Asha and Ben on the action side, Chitra and Dev on the romance side, and Esha near 0: she has no preference.

Adding the second layer (Figure 3, right) recovers the table: no entry is off by more than 0.55. The third and fourth layers carry only small details, so the data is essentially two-dimensional: one “general liking” axis and one “action or romance” axis. In the language of the book’s movie example, the $\mathbf{v}_i$ are stereotypical films and the $\mathbf{u}_i$ stereotypical viewers; each real viewer is a mix of them.

1. **In words:** the predicted rating of viewer a for film b is the sum, over the kept layers, of (singular value) times (how much a follows the pattern) times (how much b fits it).
2. **Formula:**

$$\hat{r}_{ab} = \sum_{i=1}^k \sigma_i u_{ai} v_{bi}$$

3. **Example:** for Asha and Action 1 with $k = 2$: $14.63 \times (-0.39) \times (-0.54) + 6.57 \times 0.48 \times 0.44 \approx 3.06 + 1.39 = 4.45$, against her real rating of 5.

Extra: Real ratings tables are mostly empty: a viewer has rated a handful of thousands of films. The plain SVD needs every entry. Recommenders therefore fit the same form, $\hat{r}_{ab} = \sum_i p_{ai} q_{bi}$ (a viewer vector times a film vector), by gradient descent on the known ratings only, and use it to fill in the unknown ones. This **matrix factorisation** became famous as “Funk SVD” in the Netflix Prize (2006–2009), although strictly it is not an SVD: its vectors are not forced to be orthonormal.

5. The pseudo-inverse and least squares

Key point: Invert the SVD factor by factor, leaving zero singular values at zero: $A^+ = V \Sigma^+ U^T$. Then $A^+ \mathbf{b}$ is the least-squares solution, even when the normal equation has no inverse.

5.1 Inverting the SVD

Key point: Undo each factor in reverse order: U^T for U , $1/\sigma_i$ for each σ_i , V for V^T ; where $\sigma_i = 0$ there is nothing to undo, so keep 0.

The inverse of a product is the product of the inverses in reverse order. For $A = U \Sigma V^T$, the orthogonal factors are undone by their transposes, and Σ by replacing each σ_i with $1/\sigma_i$. When some σ_i is 0, A has no inverse, but we can still invert everything that is not 0. The result is the **Moore–Penrose pseudo-inverse**, listed in the [linear algebra roadmap Note](#).

1. **In words:** transpose Σ , replace each non-zero σ_i by $1/\sigma_i$ and leave the zeros; then sandwich it between V and U^T .
2. **Formula:**

$$A^+ = V \Sigma^+ U^T = \sum_{\sigma_i > 0} \frac{1}{\sigma_i} \mathbf{v}_i \mathbf{u}_i^T$$

3. **Example:** for the rank-1 matrix C with rows $[2, 1]$ and $[4, 2]$ ($\sigma_1 = 5$, $\mathbf{u}_1 = \frac{1}{\sqrt{5}}[1, 2]$, $\mathbf{v}_1 = \frac{1}{\sqrt{5}}[2, 1]$, from the [computing the SVD Note](#), section 5):

$$C^+ = \frac{1}{5} \cdot \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \end{bmatrix} = \frac{1}{25} \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 0.08 & 0.16 \\ 0.04 & 0.08 \end{bmatrix}$$

C has no inverse (its determinant is 0), but C^+ undoes it as far as possible: it sends each output on the column space back to the matching input in the row space. For a square matrix that does have an inverse, $A^+ = A^{-1}$.

5.2 Least squares without the normal equation

Key point: When $X^T X$ has an inverse, $X^+ = (X^T X)^{-1} X^T$, so $X^+ \mathbf{y}$ is the normal-equation solution.

The [multiple linear regression maths Note](#) (section 6) found the least-squares coefficients with the normal equation $\beta = (X^T X)^{-1} X^T \mathbf{y}$. With the thin SVD $X = U \Sigma V^T$ (all $\sigma_i > 0$), $X^T X = V \Sigma^2 V^T$, and

$$(X^T X)^{-1} X^T = V \Sigma^{-2} V^T V \Sigma U^T = V \Sigma^{-1} U^T = X^+$$

1. **In words:** the least-squares coefficients are the pseudo-inverse of the design matrix times the targets.
2. **Formula:**

$$\beta = X^+ \mathbf{y}$$

3. **Example:** the four students of that Note (CGPA 6.89, 5.12, 7.82, 7.42; packages 3.26, 1.98, 3.25, 3.67), with a column of 1s for the intercept. X has singular values 13.92 and 0.30, and $X^+ \mathbf{y} = [-0.81, 0.57]$: intercept -0.81 and slope 0.57 , the same as the normal equation.

5.3 When the normal equation breaks

Key point: With a column that copies another, $X^T X$ has no inverse and the normal equation can return nonsense; the pseudo-inverse still gives a sensible answer.

The [multiple linear regression maths Note](#) (section 7) warned that $X^T X$ has no inverse when one column can be built from others. Add a third column to the student data: the percentage, computed as $9.5 \times$ CGPA. Now the columns are dependent.

- **The SVD shows it.** The singular values of the new X are 131.65, 0.30 and 2×10^{-15} : the last is zero up to rounding, so the rank is 2, not 3.
- **The normal equation fails quietly.** Here NumPy's `inv` raised no error and returned $[3.14, 0.06, 0.02]$. Its predictions for the four students are about 4.5 to 5.2, against real packages of 2 to 3.7: garbage. (On another computer it may instead raise an error.)
- **The pseudo-inverse treats the tiny singular value as 0** and returns $[-0.81, 0.0062, 0.0589]$. Its predictions are exactly those of the two-column model, because $0.0062 + 9.5 \times 0.0589 = 0.565$, the old slope. Of all coefficient vectors that fit equally well, it picks the one with the smallest length.

So the pseudo-inverse is the safe way to solve least squares. `np.linalg.lstsq`, and scikit-learn's `LinearRegression` (through SciPy's `lstsq`), use an SVD-based LAPACK routine (`gelss`) and give the same $[-0.81, 0.0062, 0.0589]$ here.

Python: The pseudo-inverse.

```
X = np.c_[np.ones(4), cgpa]
U, s, Vt = np.linalg.svd(X, full_matrices=False)
```

```

X_plus = Vt.T @ np.diag(1 / s) @ U.T      #  $V \Sigma^{-1} U^T$ 
X_plus @ y                                #  $[-0.811, 0.565]$ 
np.linalg.pinv(X) @ y                      # the same, built in
np.linalg.pinv treats singular values below a small cut-off (relative to  $\sigma_1$ ) as zero,
which is what rescues the three-column case.

```

Extra: Dropping tiny singular values is also a form of regularisation. Ridge regression, from the [ridge regression maths Note](#), has a neat SVD form: it multiplies the contribution of each direction by $\sigma_i^2 / (\sigma_i^2 + \lambda)$, which is close to 1 for large σ_i and close to 0 for tiny ones. The pseudo-inverse keeps or drops a direction completely; ridge shrinks it smoothly. This is why scikit-learn's Ridge offers `solver="svd"`.

6. Summary

ML tool	Matrix we take the SVD of	What the SVD gives
PCA	centred data X_c	PCs = rows of V^T ; variances σ_i^2/n ; projected data $U\Sigma$
Latent semantic analysis	document-word counts	topics = top $\mathbf{v}_i$; documents in topic space $U_k\Sigma_k$
Ratings / recommenders	viewer-film ratings	viewer types $\mathbf{u}_i$, film types $\mathbf{v}_i$; predictions $\sum \sigma_i u_{ai} v_{bi}$
Least squares	design matrix X	$\beta = X^+ \mathbf{y}$, safe even with dependent columns
Compression, noise reduction	any data matrix	best rank- k approximation (see the low-rank approximation Note)

- PCA is the SVD of the centred data; the covariance matrix is never needed.
- Recent scikit-learn picks between the SVD and the covariance route by the shape of the data.
- LSA links documents that share a topic but no words, by keeping the top singular directions.
- A ratings matrix's top singular vectors read as viewer types and film types.
- The pseudo-inverse inverts only the non-zero singular values; it gives least squares without the normal equation's breakdowns.

7. Key terms

Term	Meaning
PCA through the SVD	Taking the principal components from V of the centred data, with variances σ_i^2/n and scores $U\Sigma$
Randomized SVD	A fast method that finds only the top k singular vectors, used by scikit-learn for large data
Latent semantic analysis (LSA)	Describing documents by their top- k singular directions of the document-word matrix, so that texts on one topic align
TruncatedSVD	scikit-learn's rank- k SVD without centring; works on sparse matrices
Matrix factorisation (recommenders)	Predicting ratings as a viewer vector times a film vector, fitted on the known ratings only
Moore–Penrose pseudo-inverse (A^+)	$V\Sigma^+U^T$: the SVD inverted with zero singular values left at zero

Term	Meaning
Minimum-norm solution	Among all equally good least-squares solutions, the one with the smallest length; what $A^+\mathbf{b}$ returns